화학특강 (2010년 2학기)

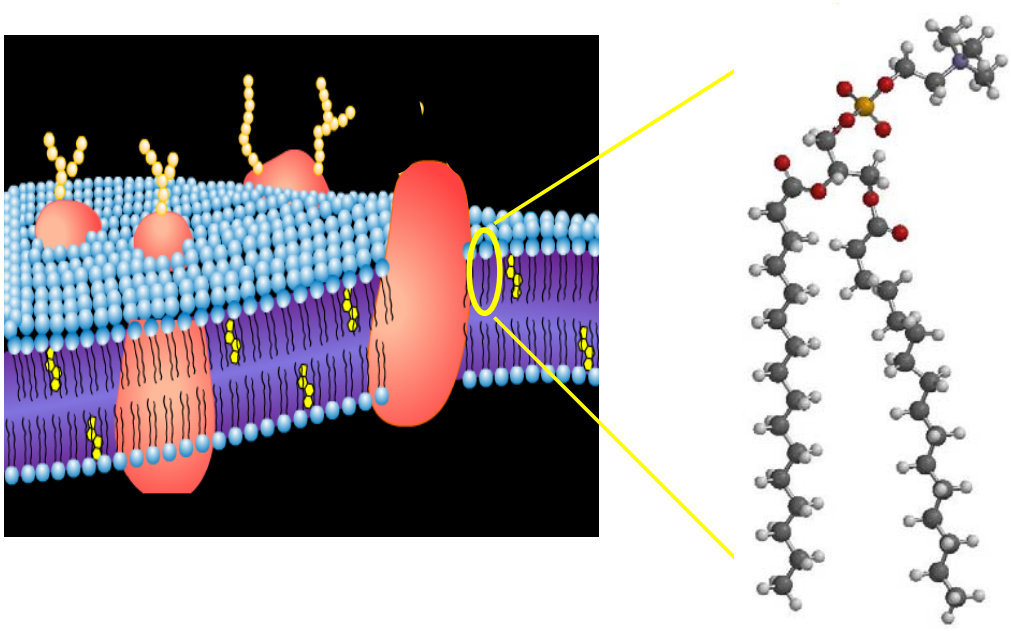
제3장 **Molecular Self-Assembly**

황현석

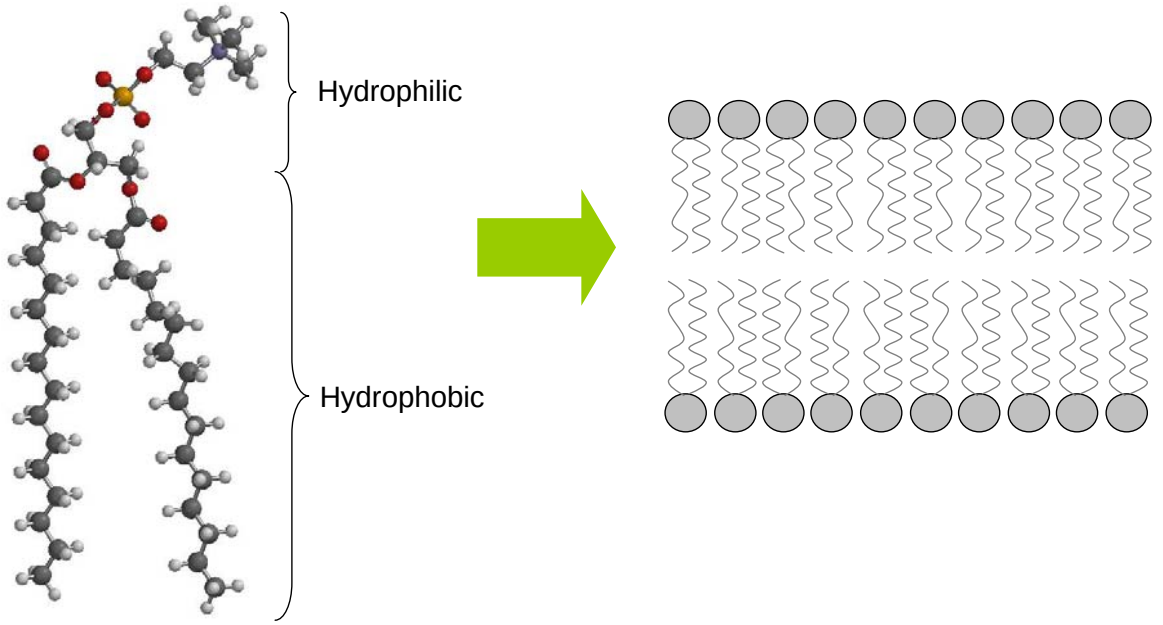
강원대학교 화학과

2010년 10월 11일 월요일

Cell Membranes and Lipid Molecules

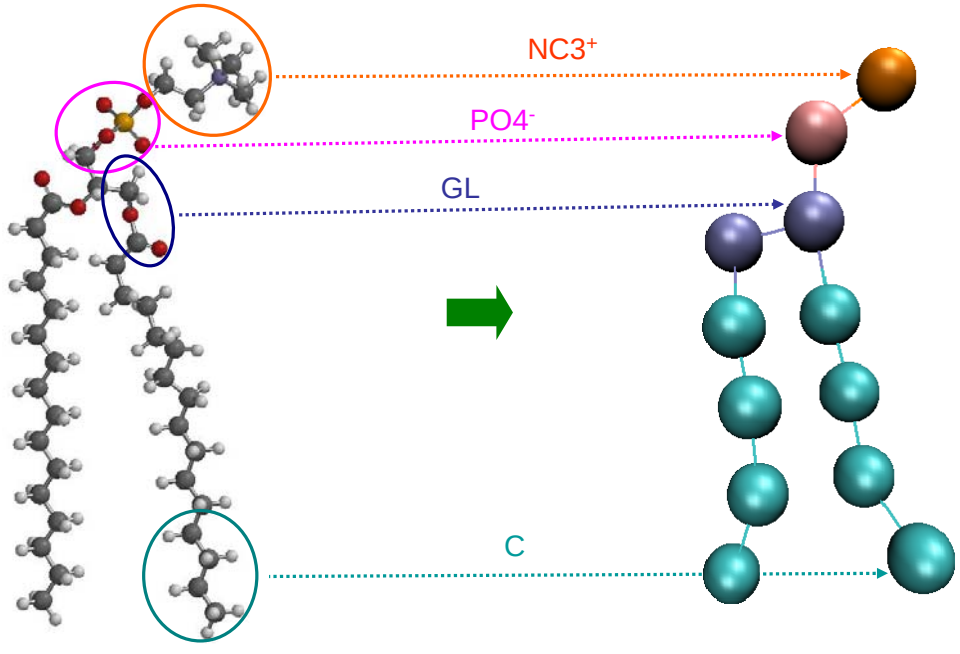


Lipid molecule as an amphiphile



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CG modeling for DPPC lipid molecules

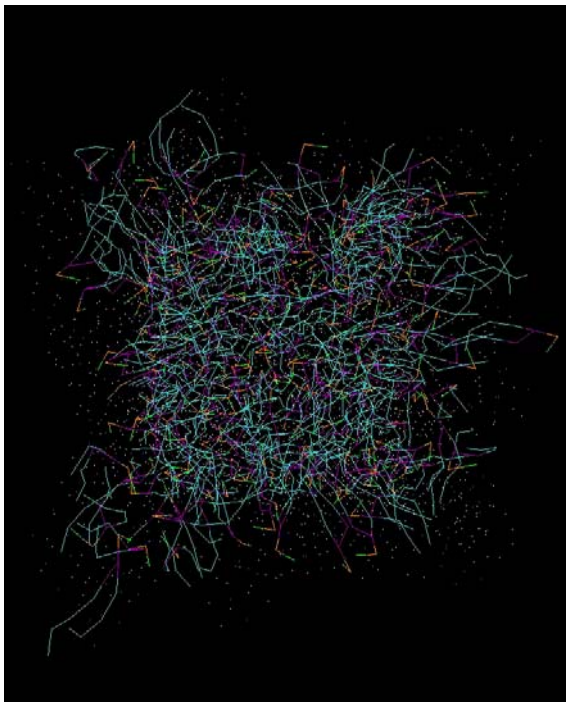


all atom model

coarse grained model

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Self-Assembly of Lipid Molecules



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Principles of Self-Assembly (1)

1. Self-assembly is a process in which components, either separate or linked, spontaneously form ordered aggregates.
2. Self-assembly can occur with components having sizes from the molecule (molecular self-assembly) to the macroscopic, provided that appropriate conditions are met.
3. In molecular self-assembly, the molecular structure determines the structure of the assembly.
4. The sucess of self-assembly in a molecular system is determined by five characteristics of the system.

   1) Components: A self-assembling system consists of a group of molecules or segments of macromolecule that interact with one another.

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Principles of Self-Assembly (2)

2) Interactions: Self-assembly occurs when molecules interact with one another through a balance of attractive and repulsive interactions. This interactions are generally weak and noncovalent (van der Waals, Coulomb interactions, hydrophobic interactions, and hydrogen bonds) interactions.

3) **Reversibility (equilibrium):** For self-assembly to generate ordered structures, the association either must be reversible or must allow the components to adjust their positions within an aggregate once it has formed.

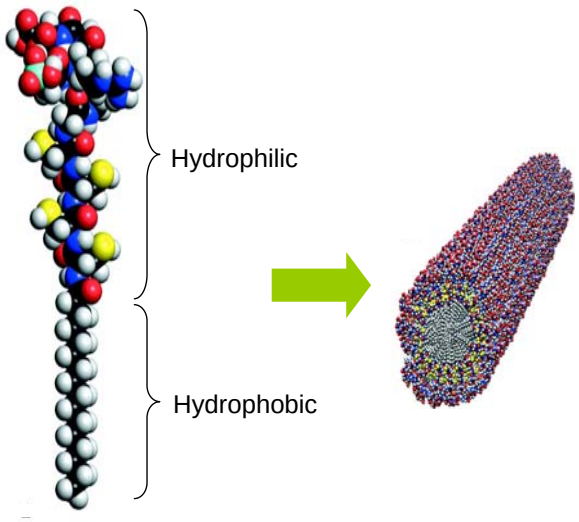
4) Environment: The self-assembly of molecules normally is carried out in solution or at an interface to allow the required motion of the components.

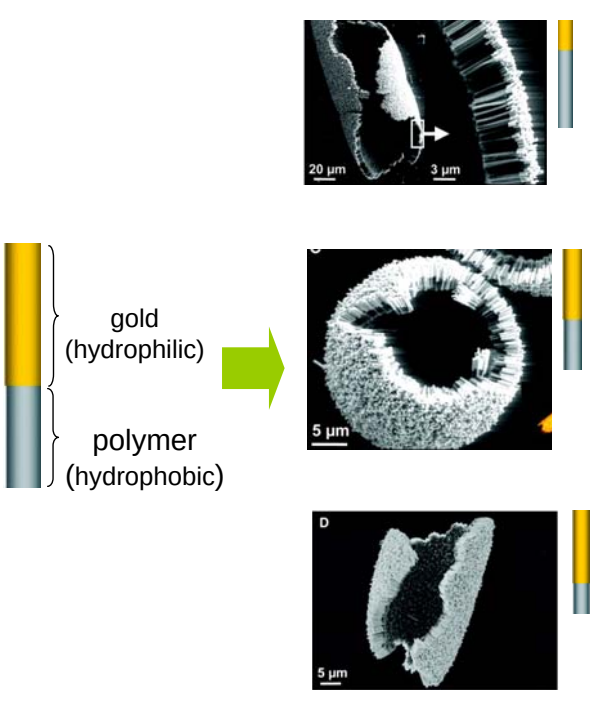
5) **Mass transport and agitation**: For self-assembly to occur, the molecules must be mobile. In solution, thermal motion provides the major part of the motion required to bring the molecules into contact.

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Principles of Self-Assembly

- Components

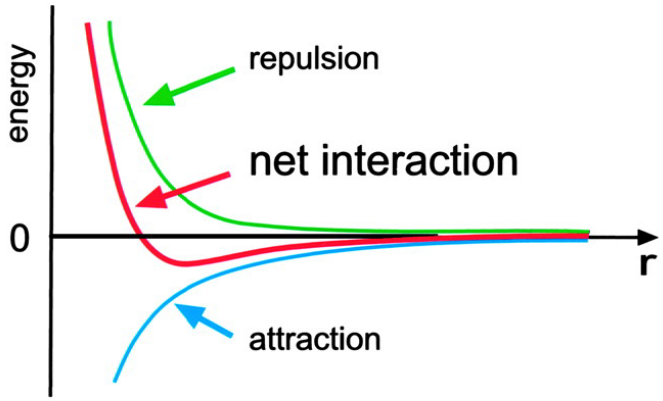


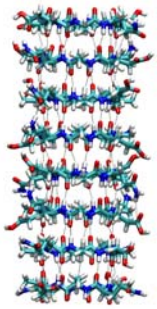


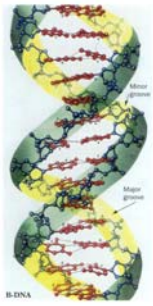
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Principles of Self-Assembly

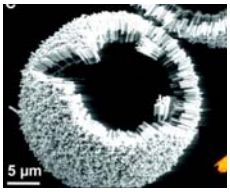
- Interactions









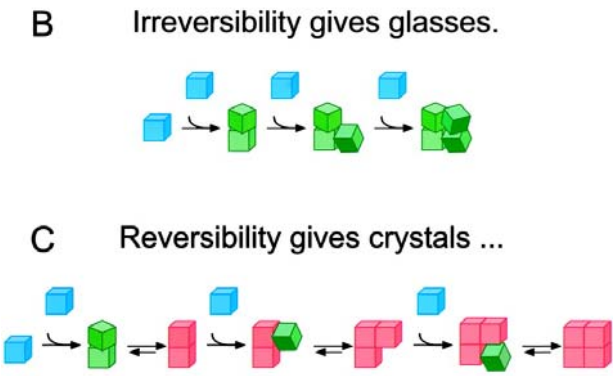


G.M Whitesdies and M. Boncheva, Proc. Nat. Acad. Sci. USA **98**, 4835 (2002)

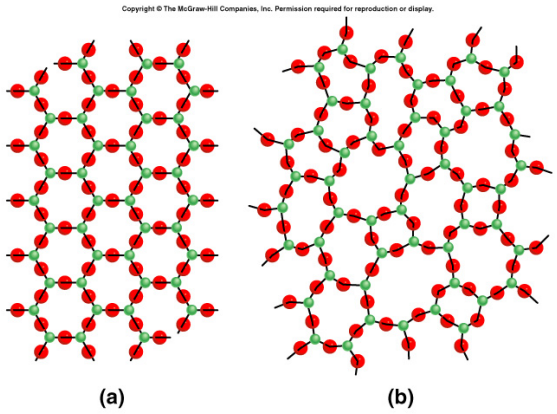
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Principles of Self-Assembly

- Reversibility (equilibrium)



Crystalline quartz (a) and noncrystalline quartz (b)

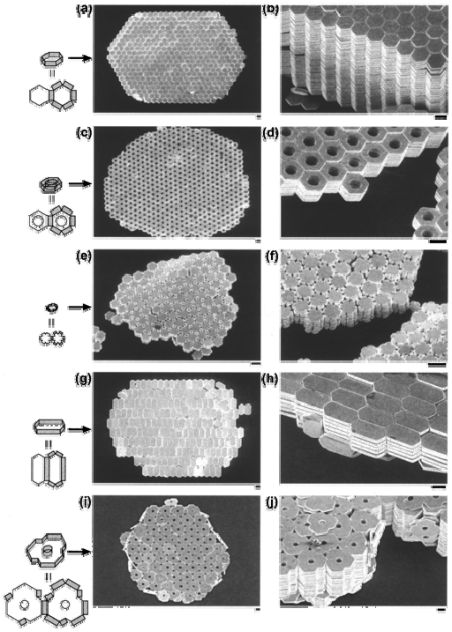


Chang, 11.7

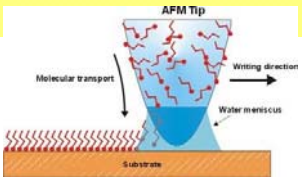
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Principles of Self-Assembly

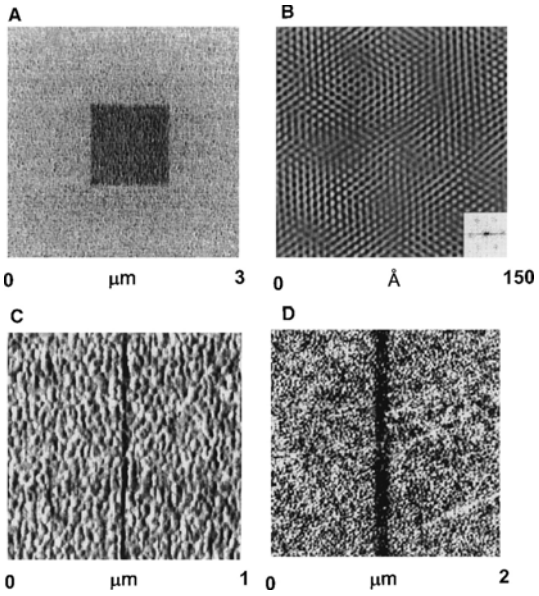
- Environment



Whitesdies et al., JACS **123**, 7677 (2001)

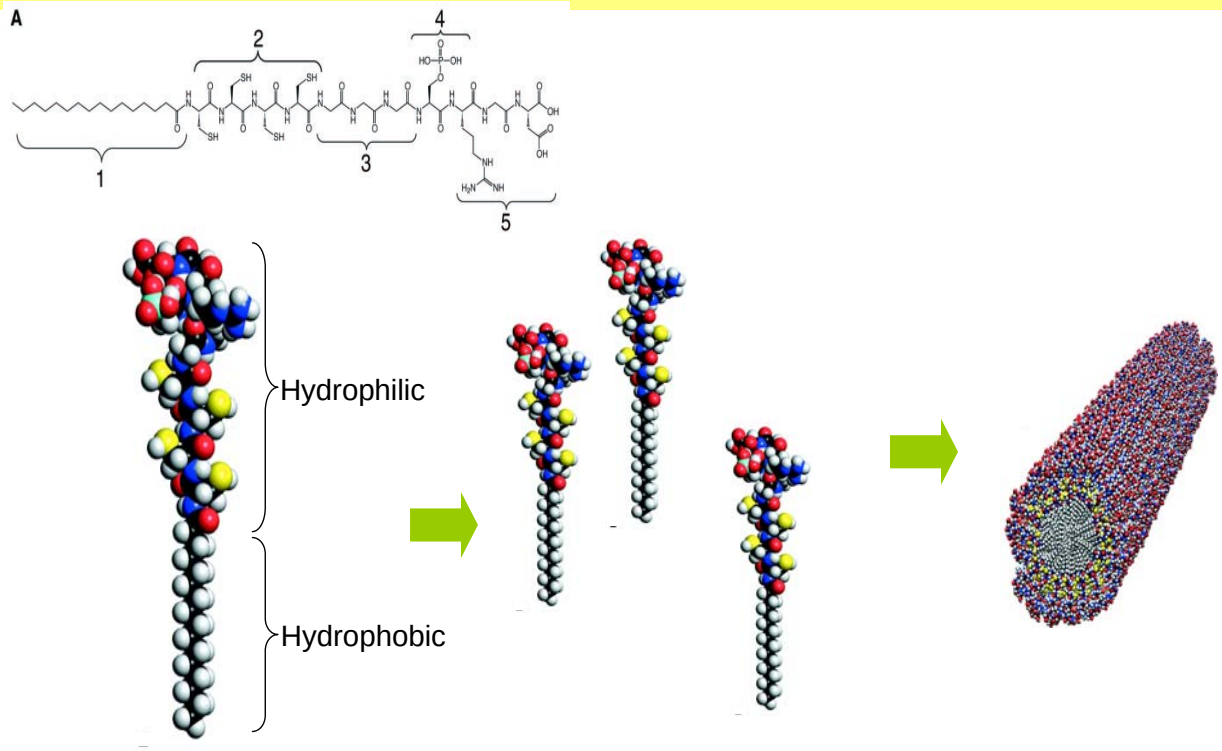


DPN on gold and on mica



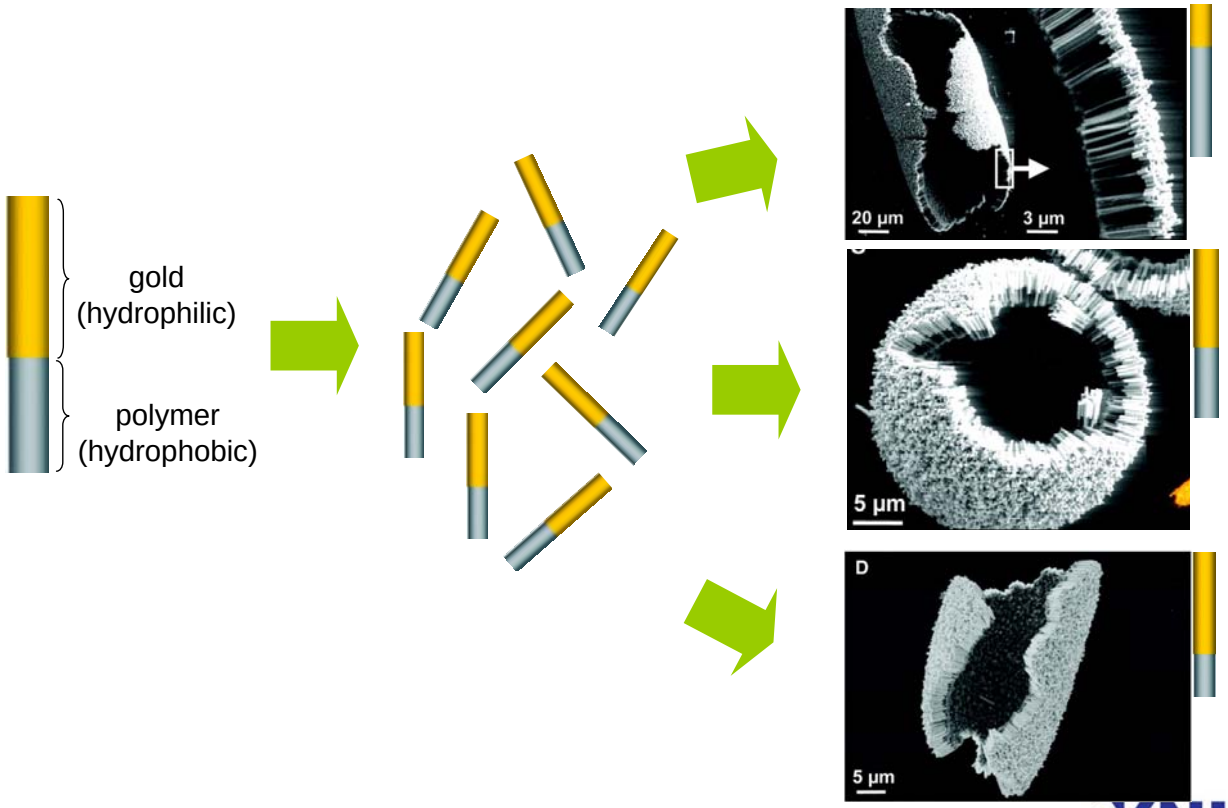
Mirkin et al., Science **283**, 661 (1999)

Self-Assembly of Peptide Amphiphile Nanofibers



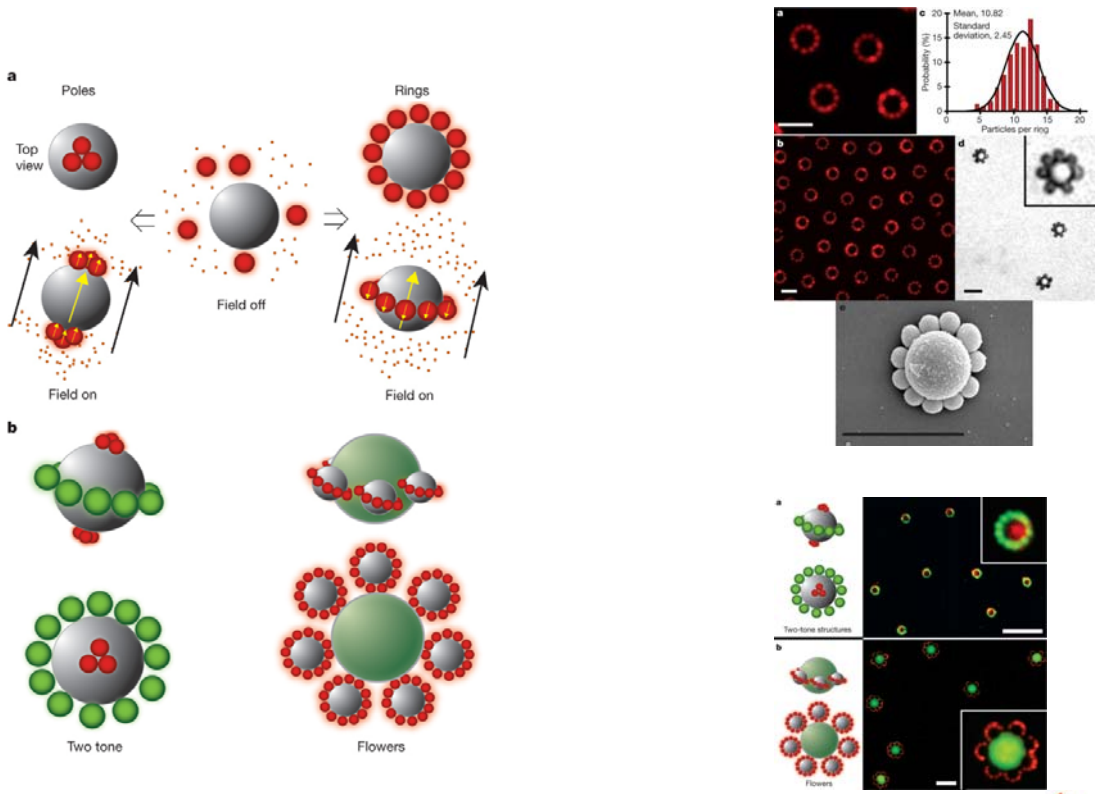
Hartgerink et al., Science **294**, 1684 (2001)

Self-assembly of metal-polymer amphiphiles



Park et al., Science **303**, 348 (2004)

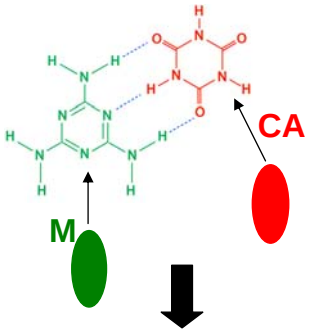
Magnetic assembly in colloidal particle mixtures

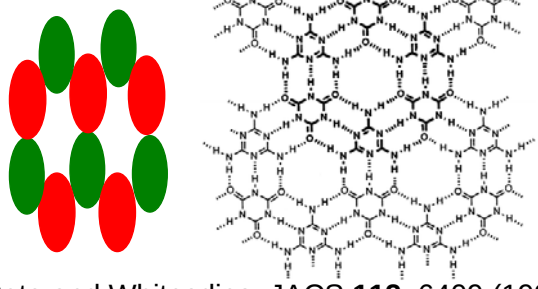


Erb et al., Nature **457**, 999 (2009)

Molecular Self-Assembly

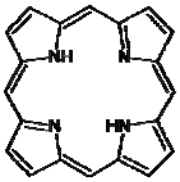
- Melamine (M) - cyanuric acid (CA) complex (hydrogen bonds)



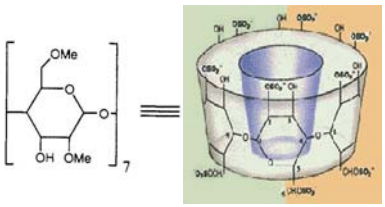


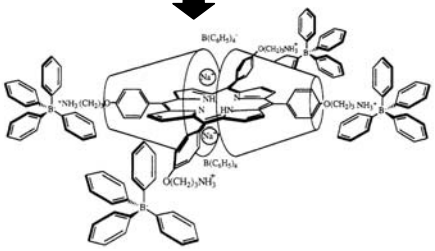
Seto and Whitesdies, JACS **112**, 6409 (1990)

- Supramlecular complex containing porphyrin and β-cyclodextrin (hydrophobic interaction)



porphyrin



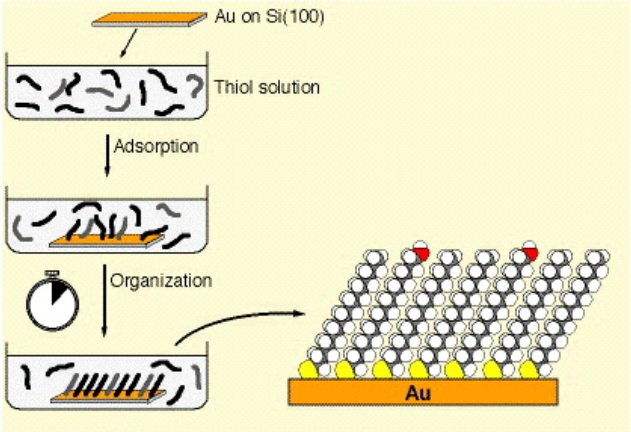


Manka and Lawrence, JACS **112**, 2440 (1990)

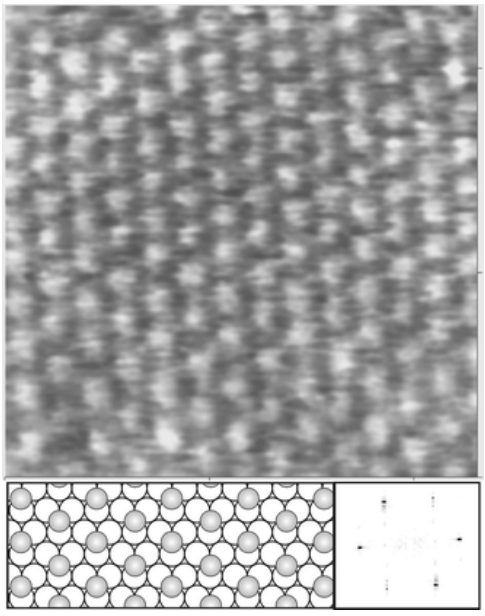
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Self-Assembled Monolayers (SAM)

- Alkanethiolates on Au



Alkanethiolates are tilted approximately 30° from the normal to the surface to maximize the van der Waals interactions between adjacent methylene group



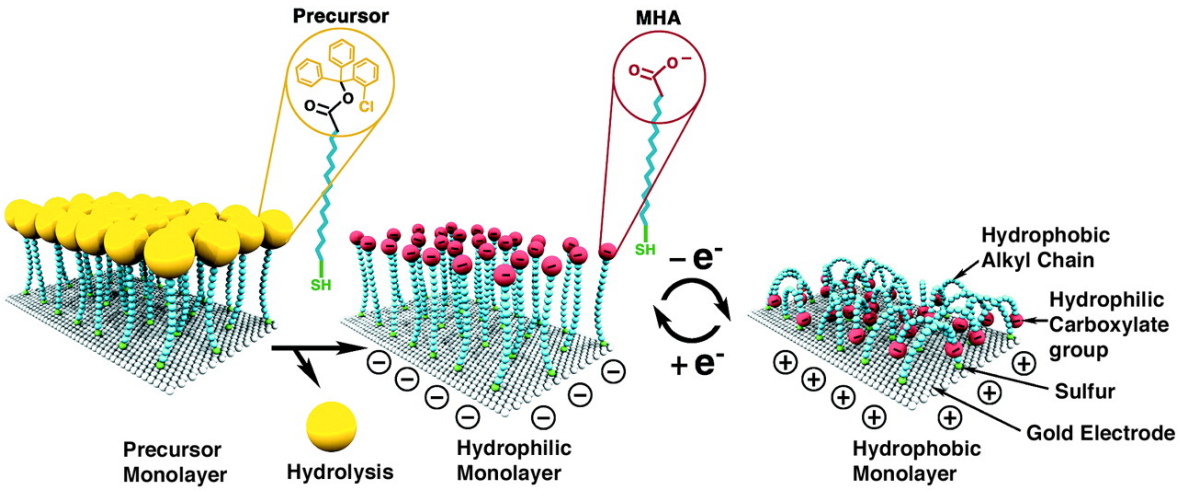
Hexanethiolates on Au[111]

Vericat et al., PCCP **7**, 3258 (2005)

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SAM Applications: Electrowettability Switch

- Surface engineering: hydrophilic monolayer ↔ hydrophobic monolayer



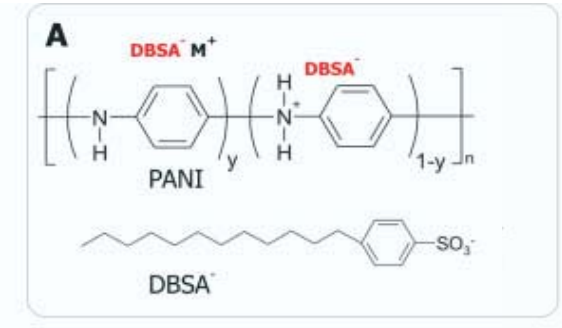
MHA = (16-mercapto)hexadecanoic acid

Lahann et al., Science **299**, 371 (2003)

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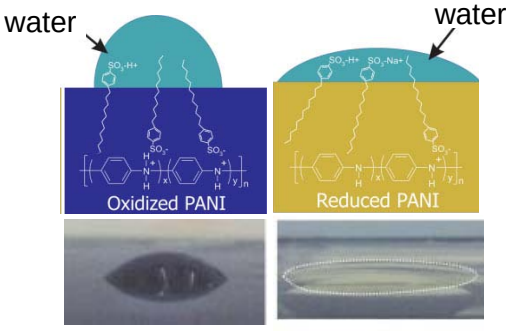
SAM Applications: Electrowettability Switch

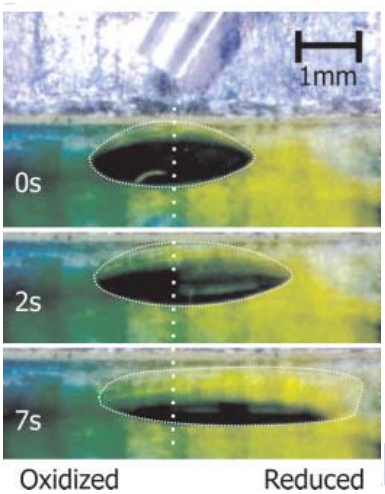
- Surface engineering: the combination of a redox-active conducting polymer and a surfactant



PANI: polyaniline (surfactant)
DBSA: dodecylbenzenesulfonic acid (conducting polymer)

Isaksson et al., Adv. Mater. **16**, 316 (2004)

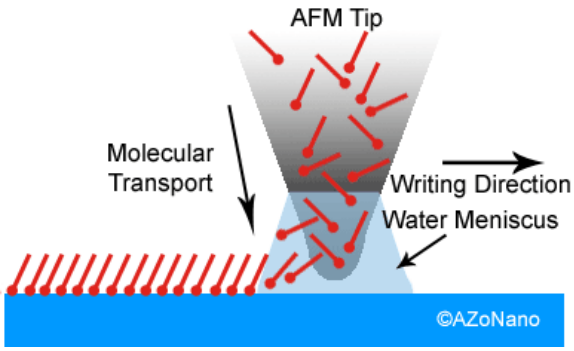


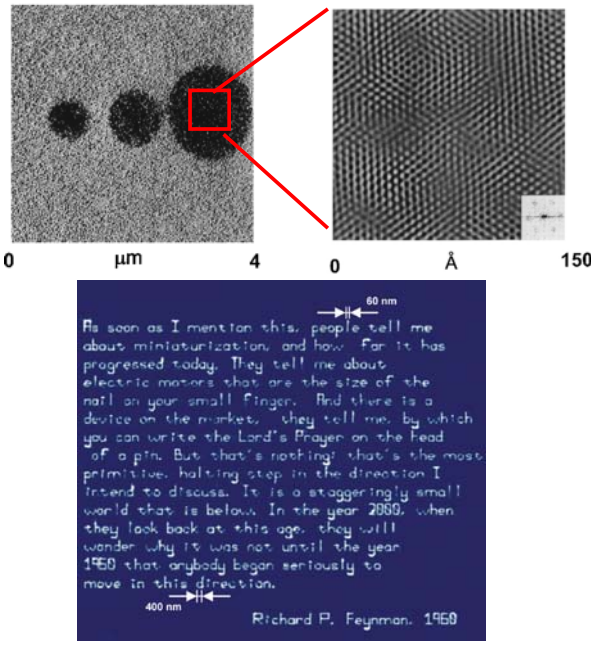


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Patterning SAM via Dip Pen Nanolithography (DPN)

- Coated AFM tip is scanned across a substrate, depositing chemical “ink” in its wake.

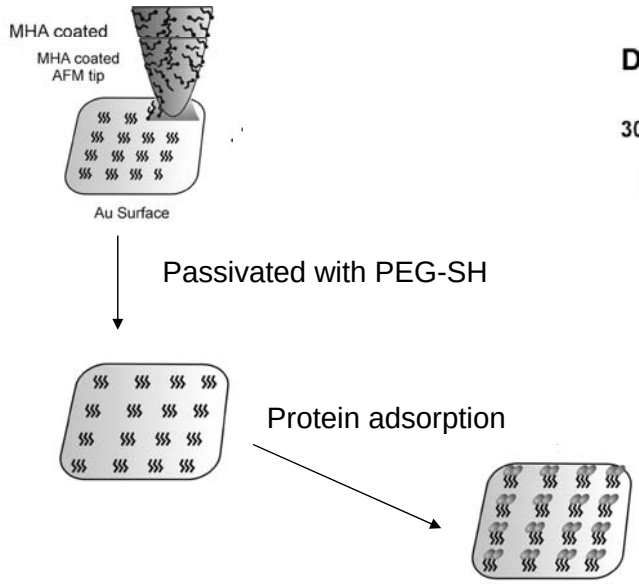


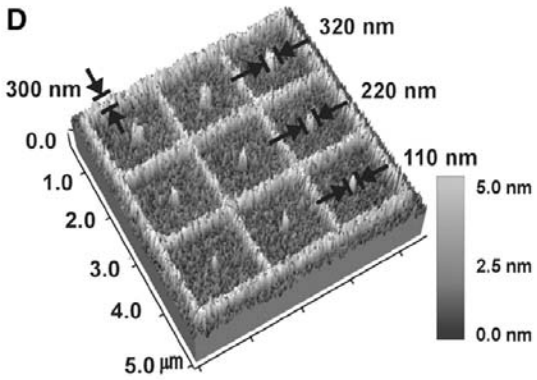


Mirkin et al., Science **283**, 661 (1999)

SAM Applications

- SAM for templates: Biomolecules

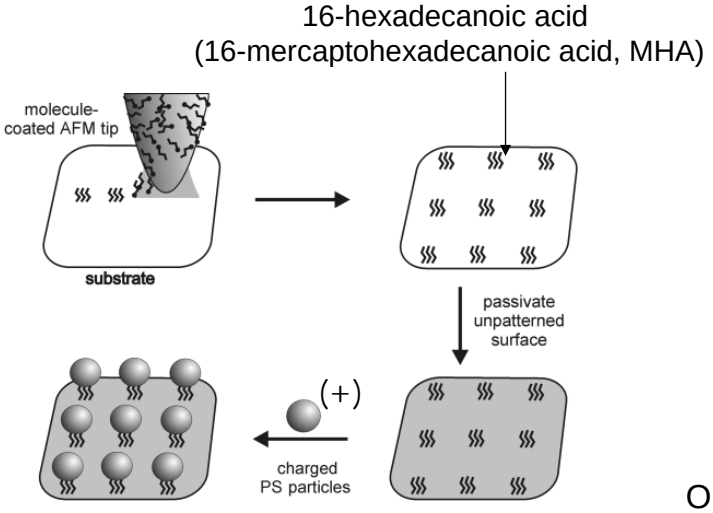


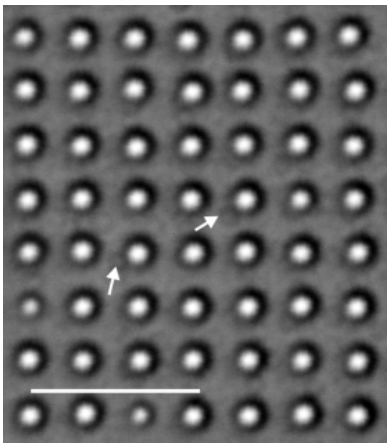


Mirkin et al., Science **295**, 1702 (2002)

SAM Applications

- SAM for templates: electrostatic assembly of charged polystyrene particles

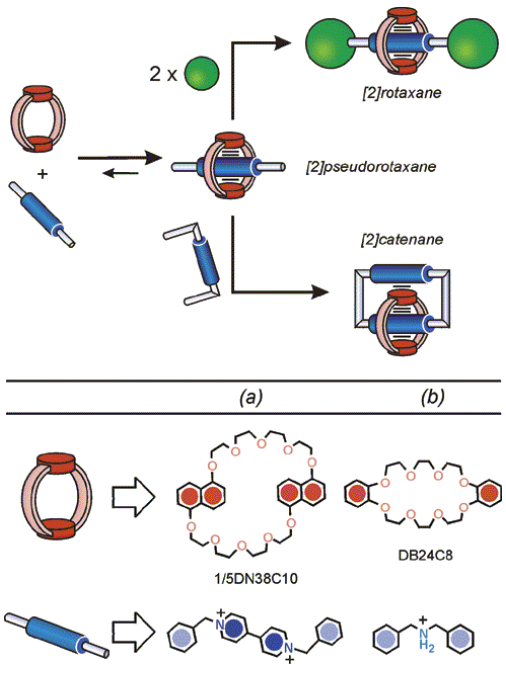




Optical micrograph of 1.0 μm amine-modified PS particles on MHA templates

Mirkin et al., Angew. Chem. Int. Ed. **40**, 3069 (2001)

Interlocked Molecules

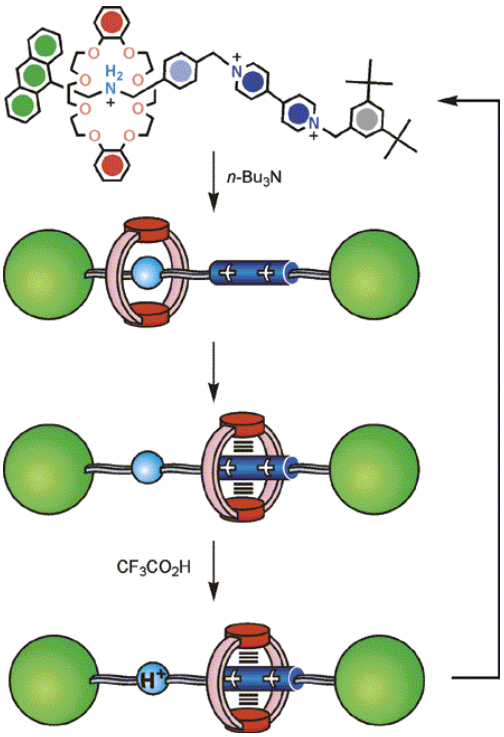


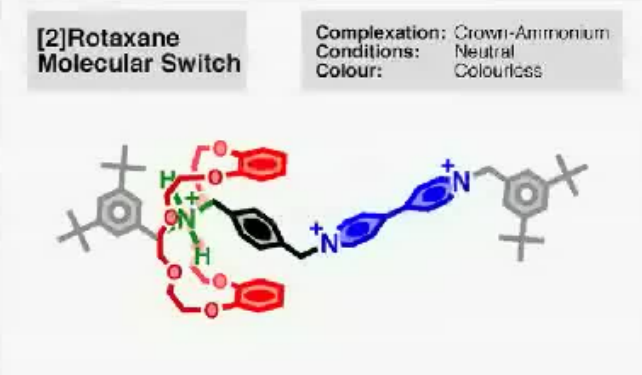
- **Supramolecular chemistry:** the area of chemistry that focuses on the noncovalent bonding interactions of molecules.
- **Rotaxane**: molecular assemblies in which a linear component is mechanically threaded through one or more cyclic molecules with the ends of the linear component terminated with bulky groups.
- **Catenane**: linked supramolecular systems consisting of cyclic subunits held by a mechanical means
- **Host guest adduct**: Interaction between π-electron rich hydroquinone and π-electron deficient pyridinium to form π stack (3.7 Å between rings). dispersive, electrostatic interactions.

Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

Rotaxanes: A Molecular Shuttle

"rot" means "wheel and axa means " axis





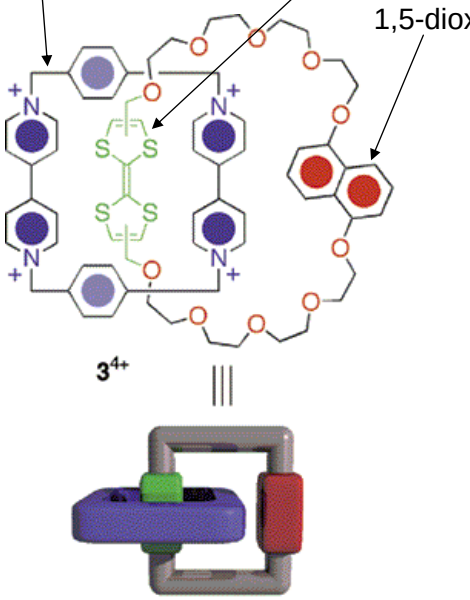
Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

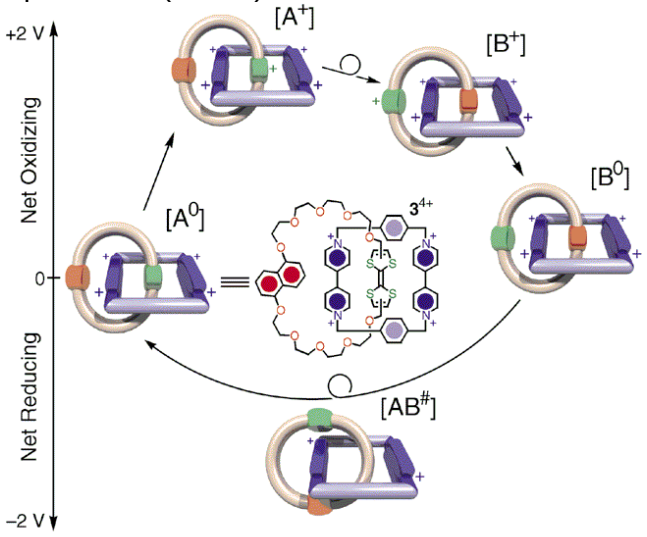
Catenanes: A Molecular Switch

cyclobis(paraquat-p-phenylene) ($CBPQT^{4+}$)

tetrathiafulvalene (TTF)

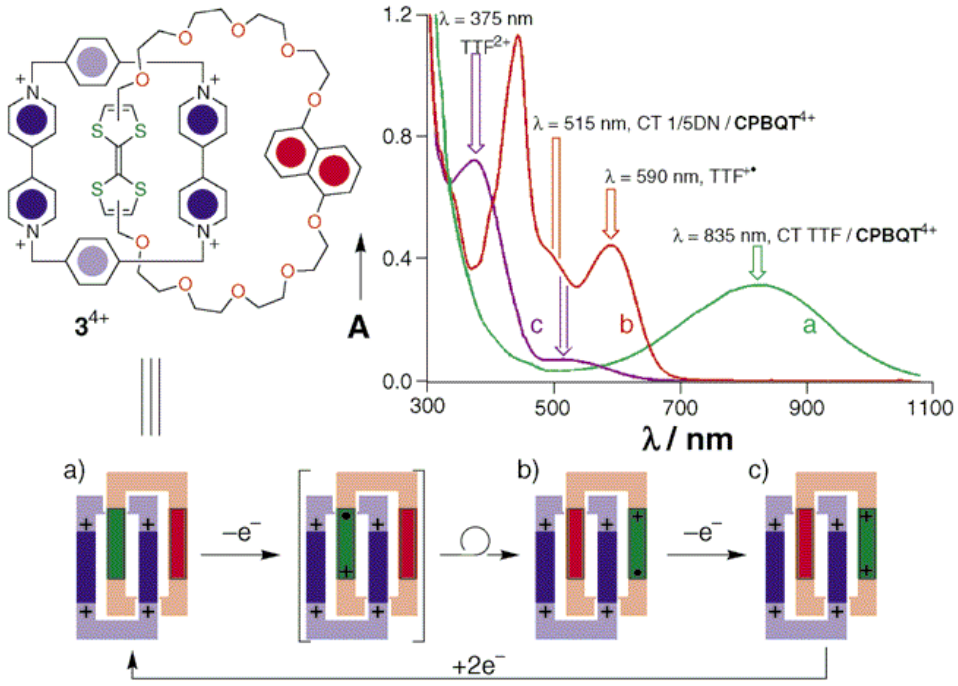
1,5-dioxynaphthalene (1/5DN)





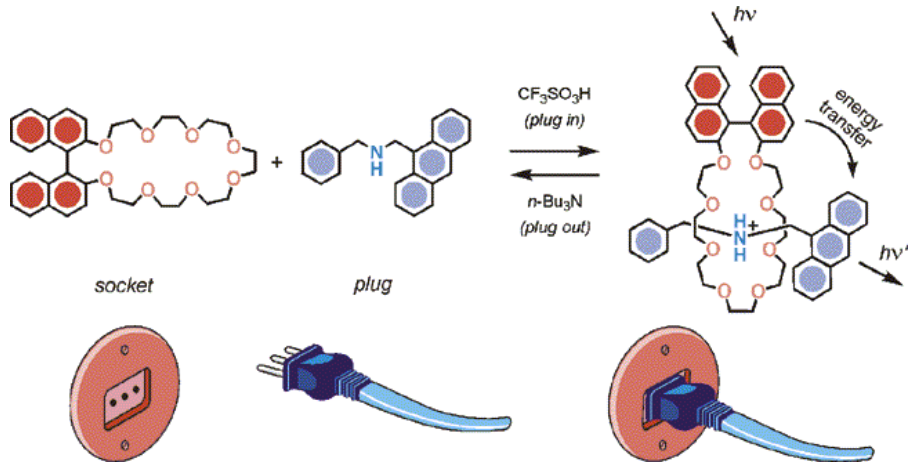
Balzani et al., Acc. Chem. Res. **34**, 445 (2001); Stoddart et al., Acc. Chem. Res. **34**, 433 (2001)

Catenanes: A Molecular Switch



Balzani et al., Acc. Chem. Res. **34**, 445 (2001); Stoddart et al., Acc. Chem. Res. **34**, 433 (2001)

A Molecular Socket-Plug device



Balzani et al., Acc. Chem. Res. **34**, 445 (2001)